CLEANQUEST · PRODUCT & ARCHITECTURE REPORT

CleanQuest — Value Proposition, Feature Tour, Technical Architecture & Post-Hackathon Roadmap

Thesis: Commercial night-shift cleaning is invisible until someone complains. CleanQuest turns a 90-second cleaning run into an AI-verified game, a 10-card daily micro-training deck, a moment-of-work sponsor channel, and an exportable geofenced inspection log owned by the cleaner.

Zero-Key Cloud + On-Device Edge AI: Every cloud scorecard and bi-weekly training season is powered by **Meta** `Muse Spark` models (`meta-models/muse-spark-open` on **Baseten** primary with automatic failover to **Fireworks AI**, plus direct **Meta Model API** support for `muse-spark-1.3`, `sam-3.1`, and `muse-voice-transcribe-1.0`), paired with on-device **Google MediaPipe** perception and **715 pre-rendered trilingual neural voice clips** (`EN` / `es_MX` / `PL`).

▶ Launch Live App | Breakroom Spectator Board | Download PDF | Submission Playbook

1 · Value Proposition & Closed-Loop Thesis

2 · Deep Technical Architecture: Models & Providers

3 · Mobile & Desktop Screen-by-Screen Feature Tour

4 · Post-Hackathon MVP Roadmap

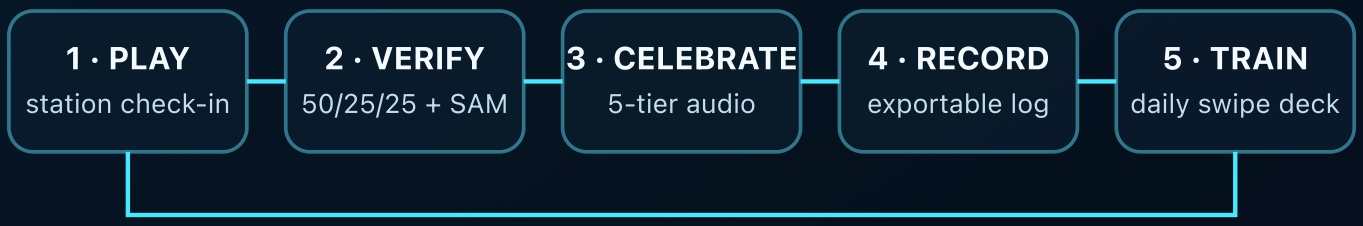
1 · Value Proposition — Who CleanQuest Serves & Why It Exists

The Core Problem: Commercial cleaners work alone at 2 a.m. across office towers, clinics, schools, and airports. When they do great work, nobody sees it; when a tenant complains the next morning, contractors burn hours driving site-to-site for re-checks, and training binders sit unopened in janitor closets.

Stakeholder	Pain Point Today	CleanQuest Value Proposition	Live Product Surface
1. The Cleaner *Visibility, Skill & Portable Proof*	Good work is invisible; language barriers and complex software block adoption; no portable proof of quality when changing jobs or shifts.	Proves visible cleaning quality in under 90 seconds with two photos, zero typing, 44px glove-friendly controls, and full trilingual voice & UI (**English, Mexican Spanish** `es_MX`**, and Polish** `PL`). Every run builds an exportable inspection record (`⬇ SAVE BACKUP` JSON + `⬇ DOWNLOAD SPREADSHEET` CSV), while the **Daily Swipe Deck** teaches pro technique one card at a time.	Welcome Hub, Consent Read-Aloud, Station Check-In, Locked 1:30 Timer, 50/25/25 Scorecard, 10-Card Daily Deck, Inspection Log Export.
2. Building Service Contractors & Facility Operators *Proof Without Re-Checks*	"They didn't clean" tenant disputes, expensive supervisor re-check drives, and subjective manual inspections.	Objective, timestamped Before/After visual evidence and per-surface AI scorecards across **10 building sectors, 13 functional areas, and 28 cleaning spots**. Scores are read-only (`The system scores, nobody edits`),	Station Taxonomy Picker, Per-Surface Scorecard + SAM Overlays, 1-Tap Dispute Panel, Geofenced Inspection Log.

		with a 1-tap worker appeal flow (`⚑ DISPUTE SCORE`) that attaches a typed or spoken note without allowing anyone to alter a score.	
3. Industry Associations & Sponsors *Training at the Moment of Work*	Paper SOP binders and classroom videos are forgotten on the shift floor; brands pay $15K–$55K at trade shows to reach crews indirectly.	Benchmark times, 8-fold microfiber technique, top-to-bottom wipe order, and surface-safe pH rules are taught right at the station and inside the **Daily Swipe Deck**. Sponsor logos appear on post-answer training cards and challenge leaderboards — never on the scorecard itself — preserving 100% scoring neutrality.	AI Loader Neutral Technique Tip, Daily Swipe Deck Post-Answer Sponsor Slot, Leaderboard Sponsor Crest.

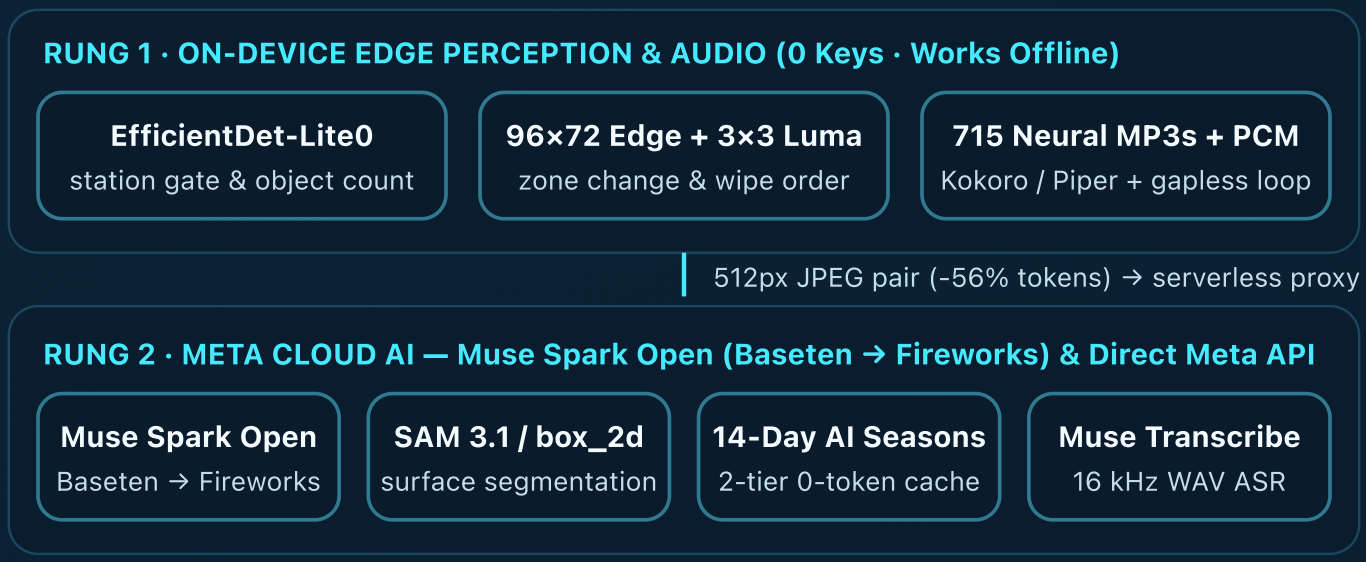
Closed-Loop Verification Thesis (Not a One-Shot Image Classifier)



e cleanliness & technique only · never claims sanitation · benchmark time is never a certification · nc

2 · Deep Technical Architecture — Models, Providers, Pipeline & Audio

Architectural Principle: CleanQuest is a zero-build, offline-capable Vanilla JS application (`index.html`, `app.js`, `styles.css`, `deck.js`) that combines **Rung 1 on-device edge perception** (instant, zero network required) with **Rung 2 cloud verification powered exclusively by Meta `Muse Spark` and `SAM 3.1` models**.



cloud model is Meta (Muse Spark / SAM / Muse ASR) · every run logged to the scrollable in-app Provider Ru

1 · Complete Models & Providers Matrix

Layer & Provider	Exact Model ID / Endpoint	Where It Runs	Ho Op

Meta (Open Weights via Baseten Primary)	`meta-models/muse-spark-open-1026`
	`meta-models/muse-spark-open`
	`inference.baseten.co/v1/chat/completions`
		Serverless Cloud Proxy (Default, 0 User Keys) or Direct Baseten Key	**Pri & s** the Be sch (C Gra f 10(ret [yr

			sur cel ger trili
Meta (Open Weights via Fireworks AI Failover)	`accounts/fireworks/models/muse-spark-open` (via serverless partner proxy)	Serverless Cloud Proxy (Automatic Failover & Dual Benchmark)	**Hig be** mc firs onl par `du` TEI in p tel `co `to **Pro**
Meta Model API (Direct Cloud)	`muse-spark-1.3` `https://api.meta.ai/v1/chat/completions`	Direct Browser → Meta API (Optional User Key)	**Dir ver** `de` gra and sc Mc
Meta Segment Anything (SAM 3.1)	`sam-3.1` `https://api.meta.ai/v1/responses`	Direct Browser → Meta API (with `box_2d` fallback)	**Su (`g` `sa` `gla` bo dra sur ph co dir `ge fall**

			co Spa

Meta Voice ASR	`muse-voice-transcribe-1.0`
	`https://api.meta.ai/v1/asr/transcribe`
		Direct Browser → Meta API (Multipart WAV)	**Sp
			no
			(** `t`
			Ca
			`Me`
			bro
			`0`
			kH
			(`v`
			a
			rec
			rav
			att
			are

Google MediaPipe (On-Device Edge Vision)	`EfficientDet-Lite0` (INT8 TFLite) `@mediapipe/tasks-vision@0.10.14`	100% On-Device in Browser (WebAssembly)	**Sta cle (** `v` Ru fra `m wr TV, co sel fra co su `b
On-Device Classical Vision Engine	`compareFrames` (`96×72`) + `analyzeSurfaceZones` (`3×3` grid)	100% On-Device in Browser (Canvas 2D)	**Lu 9-** Co (`a Sol (`s rat

			clu gra an (`t
Offline Neural TTS & Gapless WebAudio Engine	**Kokoro-82M** (`af_heart` EN) **Piper VITS** (`es_MX-claude-high` ES, `pl_PL-gosia-medium` PL)	Pre-rendered Offline → 715 Static 24 kHz MP3s + WebAudio PCM	**Ze ga** clip (ne TT sha an (`a EQ mu `da de `de sar `Au (`l but `G du the

2 · Deep Dive: The 6-Stage Verification & Scoring Pipeline

1. **Stage 1 — In-App Capture & Live 9-Zone Motion Sampling:** Photos can only be captured live through `MediaDevices.getUserMedia` (`assertInAppCaptureOnly` verifies no `Choose File No file chosen ` exists). While the locked 1:30 countdown runs, `sampleLiveRunActivity` samples a `36×27` luma grid every tick to track active wiping across the 9 zones and captures a `320×240` mid-run progress frame at the 45-second mark.
2. **Stage 2 — 7-Reason Layman Diagnostic Gate (`NOT SCORED`):** Before awarding any score, `calculateLocalScore` runs a strict pre-flight gate. If a capture is invalid or unchanged, `scorable` is set to `false` (or `noVisibleChange: true`), cloud calls are

skipped to save tokens, and the Results screen switches from `TOTAL SCORE` to **`NOT SCORED`** with one of 7 plain-language headlines (`WRONG AREA`, `PERSON IN PHOTO`, `TOO DARK`, `TOO MUCH GLARE`, `CAMERA MOVED`, `TOO FAR AWAY`, `NO CHANGE`), a one-sentence layman explanation of how to fix it, and a direct **`TAKE NEW PHOTOS`** or **`CLEAN & TAKE NEW PHOTO`** button (while hiding the 3 score bars, rank badge, and dispute panel).

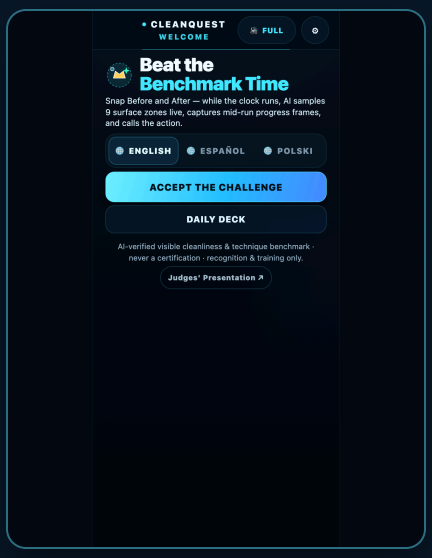
3. **Stage 3 — 50 / 25 / 25 Weighted Score Formula (`computeWeightedRunScore`):**
   - **Visible Cleanliness (50% weight):** Combines 9-zone surface improvement (`surfaceClearSignal`), zone coverage (`improvedZones / 9`), and object clearance (`clearedCount / beforeCount`), reconciled with the cloud `Muse Spark Open` per-surface scorecard when verified.
   - **Technique (25% weight):** Measures top-to-bottom wipe order (`topOrderScore` across the top row vs. bottom row of the `3×3` grid) and even zone coverage (`zoneUniformityScore`).
   - **Time vs. Benchmark (25% weight):** Awards `100%` within the 1:30 benchmark (`85–100%` of standard time), with a smooth linear deduction only when overtime.
4. **Stage 4 — Hushed-Directive Gate & 5-Tier Audio Choreography (`announceGameVerdict`):** If a run is unscorable or carries a surface directive (e.g., an untouched zone or remaining item), the `hushed` gate suppresses rank badges, confetti, and celebratory voice lines so the UI never praises an incomplete clean. On clean scorable runs, CleanQuest rolls the score counter (`0% → N%`), fires a 5-tier WebAudio musical stinger (`S-Rank ≥90%`, `A-Rank ≥80%`, `B-Rank ≥70%`, `C-Rank`, `D-Rank`), speaks the exact percentage via crossfaded `num-*` + `glue-00` clips, and chains the matching host reaction clip.
5. **Stage 5 — Downscaled Cloud Verification & Surface Overlays (`getVerifiedScore` + `getSamOverlays`):** Both Before and After captures are downscaled on-device to `512px` max edge JPEG (`0.72` quality), cutting base64 payload size and vision token cost by **~56%**. Returned `box_2d` coordinates (or `sam-3.1` masks) are rendered onto both the AFTER evidence card and the interactive Side-by-Side / Shadow comparison lightbox.
6. **Stage 6 — Two-Tier Zero-Token Bi-Weekly Season Engine (`generateLiveAiBonusCard`):** Every 14-day window (`Math.floor(Date.now() / (14 * 86400000))`), opening the Daily Deck checks Tier 1 (`localStorage`) and Tier 2 (serverless `ScriptProperties` shared cache) before calling `Muse Spark Open`. One

batched prompt generates a localized Season Theme + 3 trilingual (`EN/ES/PL`) cards, and every subsequent play across the 2-week season costs **0 tokens**.

3 · Complete Screen-by-Screen Feature Tour (Mobile 390×844 & Desktop 1440×900)

Live Parity: Every screenshot below was captured directly from the current live build at **Mobile (`390×844`)** and **Desktop (`1440×900`)** viewports.

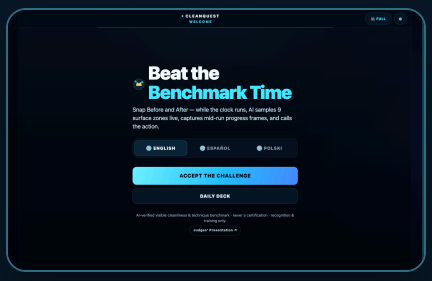
Part A · Trilingual Onboarding, Verbatim Consent Gate & Judge Demo



MOBILE — ENGLISH (390×844)



MOBILE — ESPAÑOL (390×844)

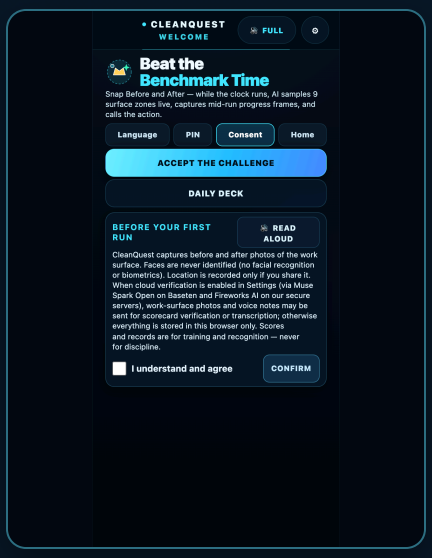


DESKTOP (1440×900)

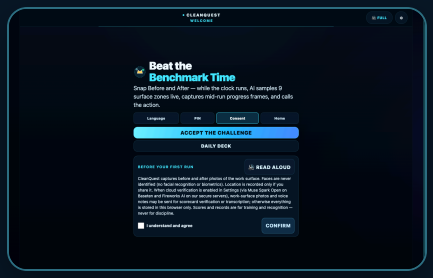
1 · Welcome Hub & 1-Tap Trilingual Switcher (`EN / ES / PL`)

1. **On-Screen Functionality:** Instant 1-tap language selector (`🌐 ENGLISH` , `🌐 ESPAÑOL` , `🌐 POLSKI`), primary `ACCEPT THE CHALLENGE` CTA, `DAILY DECK` entry, and clear benchmark disclaimer (*"AI-verified visible cleanliness & technique benchmark · never a certification · recognition & training only"*).
2. **Value Prop:** Non-technical cleaners start in their native language in a single tap with zero typing, accounts, or menus.
3. **Tech Under the Hood:** Persists language choice in `localStorage` , re-renders all `[data-i18n]` DOM nodes in place, and plays the matching pre-rendered 24 kHz neural welcome intro (`assets/audio/{en,es,pl}/welcome-intro.mp3`) on first gesture while keeping the welcome background silent.

LIVE · TRILINGUAL HUB



MOBILE (390×844)

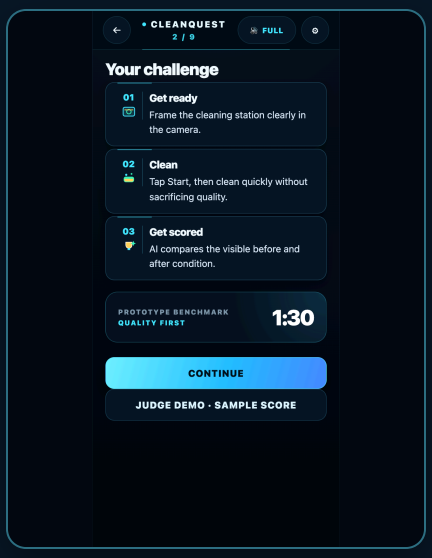


DESKTOP (1440×900)

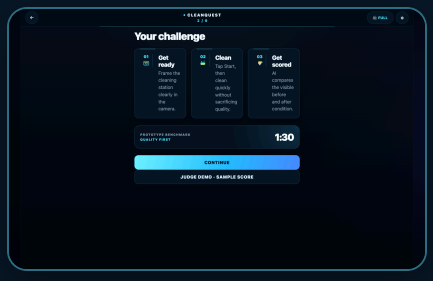
2 · First-Run Consent Gate & Verbatim Neural Read-Aloud

1. **On-Screen Functionality:** Blocks first-time camera entry until the cleaner checks *"I understand and agree"* and taps `CONFIRM`, with a prominent `READ ALOUD` button.
2. **Value Prop:** Establishes explicit worker trust upfront: work-surface photos only, faces are never identified (no facial recognition or biometrics), location is recorded only if shared, and scores are never used for discipline.
3. **Tech Under the Hood:** Stores language-scoped consent in `cleanquest.consent.v1` (switching languages re-presents consent in that language) and plays `assets/audio/{en,es,pl}/consent-00.mp3` rendered verbatim from the exact on-screen `consentBody` string.

LIVE · WORKER TRUST GATE



MOBILE (390×844)



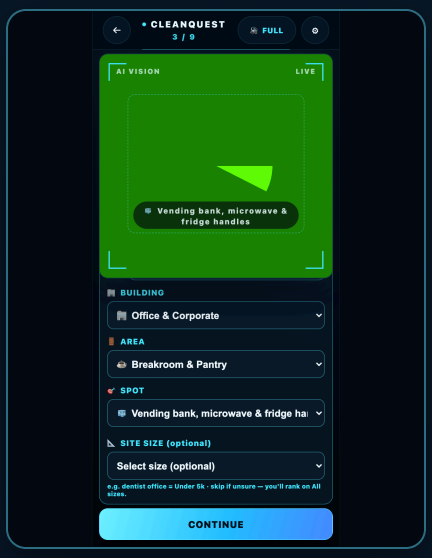
DESKTOP (1440×900)

3 · 3-Step Challenge Briefing & 1-Tap Judge Demo

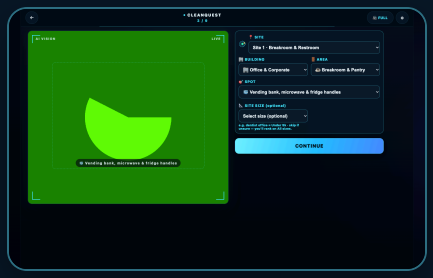
1. **On-Screen Functionality:** Concise 3-step visual summary (`01 Get ready → 02 Clean → 03 Get scored`), `1:30` prototype benchmark card, primary `CONTINUE` CTA, and 1-tap `JUDGE DEMO · SAMPLE SCORE` .
2. **Value Prop:** Lets first-time cleaners grasp the rules in 5 seconds and lets evaluators experience the full AI loader + 93% S-Rank reveal in one tap without needing a physical sink.
3. **Tech Under the Hood:** Auto-collapses on subsequent runs once `cleanquest.instructionsSeen` is set; `runJudgeDemo()` routes sample Before/After illustrations through the animated loader and 5-tier audio fanfare while marking the run `judgeDemo: true` so it never pollutes the competitive leaderboard.

LIVE · 10-SECOND DEMO PATH

Part B · Commercial Station Check-In, Locked Timer, AI Loader & Scorecards



MOBILE (390×844)

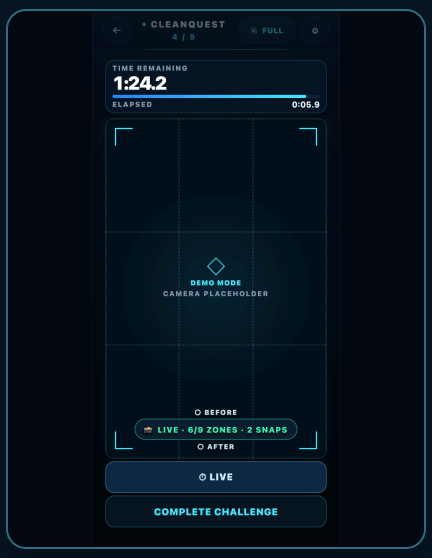


DESKTOP (1440×900)

4 · Commercial Station Check-In & Voice Explainer

1. **On-Screen Functionality:** First-visit *"How the Challenge Works"* modal with `📷 ALLOW CAMERA & START`, live camera viewfinder with framing reticle, and dropdown selectors for **Site, Building (10 sectors), Area (13 templates), Spot (28 stations), and optional Site Size**.
2. **Value Prop:** Adapts scoring weights to the exact commercial surface being cleaned (e.g., *Sink & counter* vs. *Mirror, glass & handles*) with zero typing.
3. **Tech Under the Hood:** Plays pre-rendered `camera-explain.mp3` on first visit, binds `MediaDevices.getUserMedia` (`environment` facing camera preferred), and loads station-specific scoring profiles (`SPOT_LOGIC_PROFILES`).

LIVE · 10 SECTORS · 28 SPOTS



MOBILE (390×844)

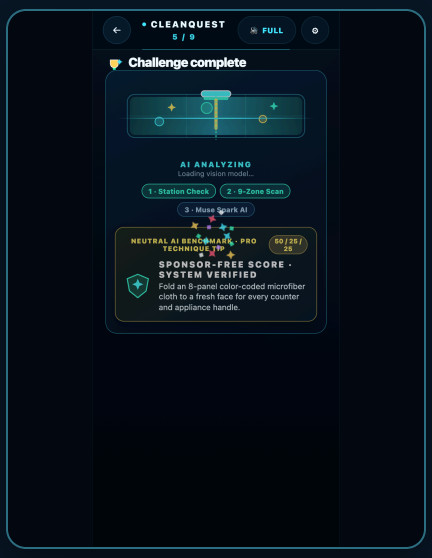


DESKTOP (1440×900)

5 · Locked 1:30 Countdown, Live 9-Zone Sampler & Challenge Track

1. **On-Screen Functionality:** Locked `1:30` countdown + elapsed bar, live `3×3` zone grid overlay (`📸 LIVE · ZONES · 2 SNAPS`), Before/After capture indicators, optional Before-shadow ghost toggle, and 1-tap `COMPLETE CHALLENGE`.
2. **Value Prop:** Prevents clock-pausing cheats, gives live visual feedback as zones are wiped, and auto-captures the After photo when `COMPLETE CHALLENGE` is tapped.
3. **Tech Under the Hood:** Loops `assets/audio/time-trial.mp3` gaplessly via a decoded WebAudio `AudioBufferSourceNode`, ducks music volume from `0.5` to `0.12` whenever 6–8s host hype clips fire, and samples `36×27` luma frames in real time.

LIVE · GAPLESS AUDIO + 9-ZONE AR



MOBILE (390×844)

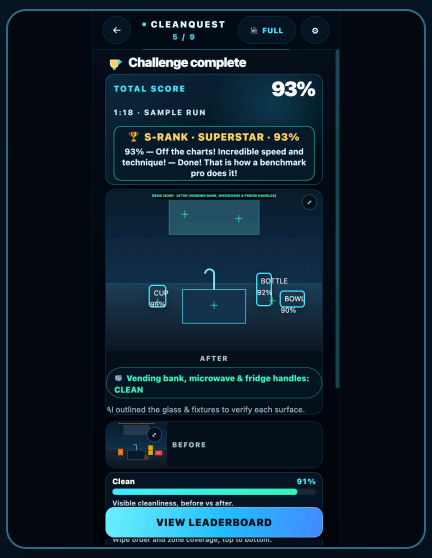


DESKTOP (1440×900)

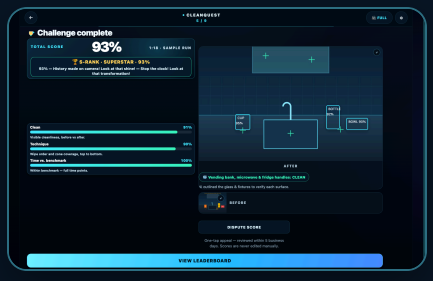
6 · Animated Squeegee/Bubble AI Loader & Neutral Technique Tip

1. **On-Screen Functionality:** Animated vector squeegee pass with rising bubbles, 3-step pipeline pills (`1 · Station Check → 2 · 9-Zone Scan → 3 · Muse Spark AI`), and a **100% sponsor-free** station-matched technique tip.
2. **Value Prop:** Turns the 1.5-second vision analysis into a Moment-of-Work micro-lesson while keeping the scoring flow strictly brand-neutral.
3. **Tech Under the Hood:** Runs MediaPipe `EfficientDet-Lite0` object detection, `96×72` frame comparison, and the serverless `Muse Spark Open` cloud request concurrently while syncing localized tips to `state.checkin.areaKey`.

LIVE · PARALLEL EDGE + CLOUD



MOBILE — SCORED RUN (390×844)

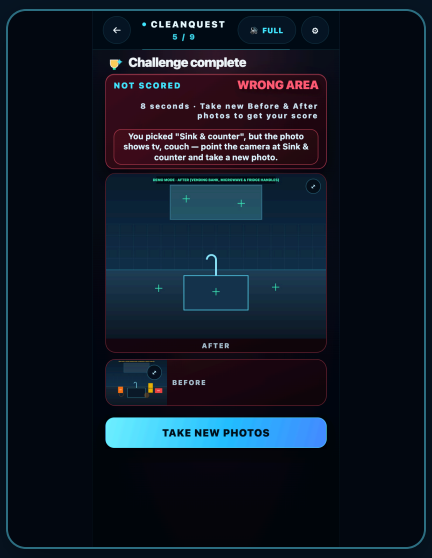


DESKTOP — SCORED RUN (1440×900)

7 · 50/25/25 Verified Scorecard, SAM Overlays & 1-Tap Dispute

1. **On-Screen Functionality:** Weighted `TOTAL SCORE` hero (`93%`), 5-tier rank badge (`🏆 S-RANK · SUPERSTAR · 93%`) + spoken transcript, maximized `AFTER` photo with glowing object/surface boxes + compact `BEFORE` reference (both tap-to-zoom into the comparison lightbox), plain-English `Clean` / `Technique` / `Time vs. benchmark` bars, and `⚑ DISPUTE SCORE`.
2. **Value Prop:** Explains every point in plain words (e.g., *"Wipe order and zone coverage, top to bottom"*) and gives cleaners a fair, immutable appeal record.
3. **Tech Under the Hood:** Combines local + `Muse Spark Open` surface grades, draws `SAM 3.1 / box_2d` overlays, and chains WebAudio stingers with crossfaded Kokoro/Piper number & reaction clips.

LIVE · 50/25/25 + SAM OVERLAYS



MOBILE — NOT SCORED · WRONG AREA (390×844)



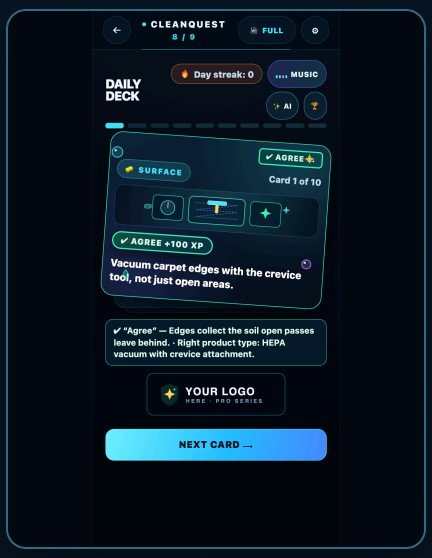
DESKTOP — NOT SCORED · WRONG AREA (1440×900)

8 · Layman `NOT SCORED` Diagnostic Screen (7 Plain-Language Reasons)

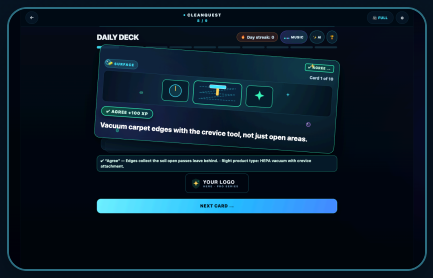
1. **On-Screen Functionality:** When a photo cannot be scored, the hero switches from `TOTAL SCORE` to **`NOT SCORED`** with a 2–3 word plain reason (`WRONG AREA`, `PERSON IN PHOTO`, `TOO DARK`, `TOO MUCH GLARE`, `CAMERA MOVED`, `TOO FAR AWAY`, `NO CHANGE`), one clear sentence explaining how to fix it (zero AI/technical jargon), and a **`TAKE NEW PHOTOS`** or **`CLEAN & TAKE NEW PHOTO`** button.
2. **Value Prop:** Non-technical users never see a confusing 0% score, a contradictory rank badge, or vague "REDO/RESCAN" labels — they know immediately why the photo didn't count and how to retake it.
3. **Tech Under the Hood:** `verifyStationEnvironment` + luminance/framing gates set `scorable: false` and `scoreDirectiveKey`, hiding `#scoreBreakdownCard`, `#verdictSummaryBadge`, and `#disputePanel` while routing the primary CTA directly back to the camera.

LIVE · LAYMAN ERROR RECOVERY

Part C · Daily Swipe Deck, Dual Leaderboards, Inspection Log & Provider Telemetry



MOBILE (390×844)

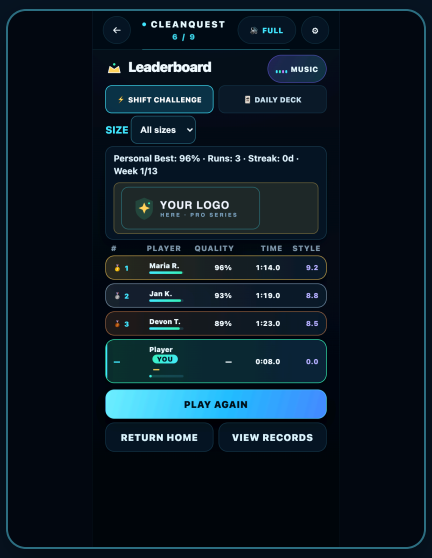


DESKTOP (1440×900)

9 · Daily Swipe Deck: Contextual SVG Art, Instant Tap-Advance & Sponsor Slot

1. **On-Screen Functionality:** 10-card daily round with 10 progress pips, contextual vector illustration matched to each question's topic, `← ✘ DISAGREE` / `✔ AGREE →` buttons & drag gestures, `✔ AGREE +100 XP` verdict stamp, plain-language explanation + recommended product type, and the `YOUR LOGO HERE` sponsor slot + `NEXT CARD →` all visible **above the fold**.
2. **Value Prop:** Teaches real technique in 60 seconds a day while keeping the sponsor placement 100% visible above the fold on mobile phones.
3. **Tech Under the Hood:** `deckMotifFor()` selects among 6 custom SVG dioramas (`product`, `cloth`, `dwell`, `technique`, `fixture`, `surface`); answering plays `deck-correct.mp3` / `deck-wrong.mp3` (`Correct!` / `Incorrect!`) over the gapless `daily-deck.mp3` loop; tapping the answered card or feedback advances immediately.

LIVE · ABOVE-THE-FOLD SPONSOR AD



MOBILE BOARD (390×844)



SPECTATOR TV (?SPECTATOR=1)

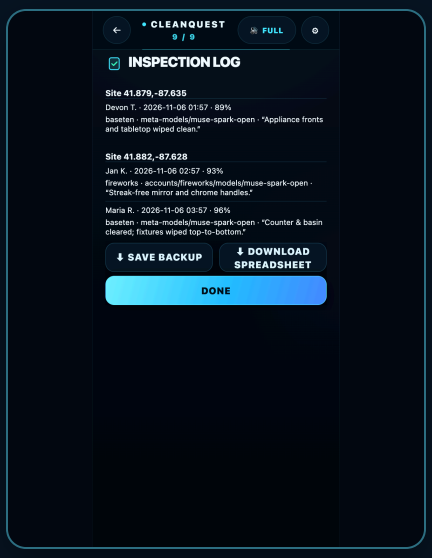


DESKTOP (1440×900)

10 · Dual-Mode Leaderboard (`⚡ SHIFT` / `🃏 DECK`), Size Filter & Spectator TV

1. **On-Screen Functionality:** 1-tap toggle between `⚡ SHIFT CHALLENGE` and `🃏 DAILY DECK` standings, facility sq-ft size filter (`All sizes`, `Under 5k`, `5k–10k`, `10k–50k`, `50k+`), personal best summary + sponsor crest, top-10 ranked rows, and `?spectator=1` breakroom display mode.
2. **Value Prop:** Recognizes top performers fairly within comparable facility sizes and never displays a bottom/shame list.
3. **Tech Under the Hood:** Sorts eligible runs by weighted 50/25/25 score (tie-breaking on faster time), automatically excludes demo/practice runs, and auto-refreshes `?spectator=1` every 15 seconds.

LIVE · TOP-10 FAIR STANDINGS



MOBILE (390×844)

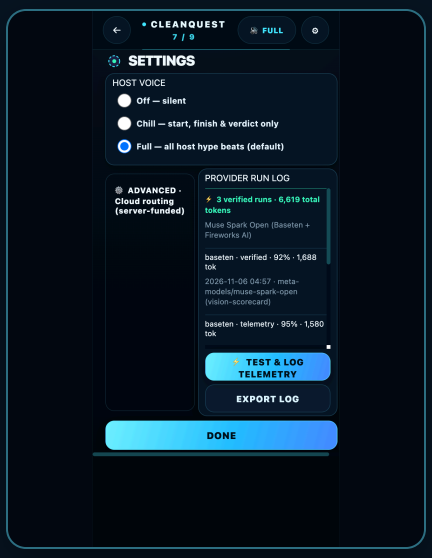


DESKTOP (1440×900)

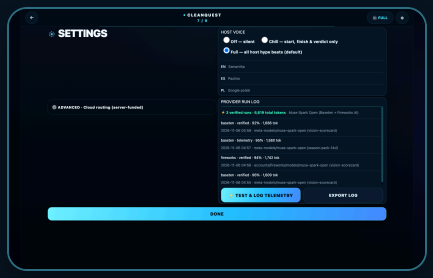
11 · Geofenced Inspection Log & 1-Tap JSON / CSV Export

1. **On-Screen Functionality:** Scrollable inspection log grouped by site coordinates (`Site 41.882, -87.628` or `Location not shared`), showing cleaner alias, timestamp, cleanliness score, AI provider/model provenance, and transcribed note — plus 1-tap `⬇ SAVE BACKUP` (JSON) and `⬇ DOWNLOAD SPREADSHEET` (CSV).
2. **Value Prop:** Gives both the cleaner and the contractor an audit-ready, portable work record with zero paperwork.
3. **Tech Under the Hood:** `recordInspection()` stamps each completed run into `cleanquest.inspection-log` (retaining 90 days via `pruneExpiredRecords`), and `buildInspectionCsv()` generates properly RFC-4180-quoted CSV spreadsheets in-browser.

LIVE · PORTABLE WORK PROOF



MOBILE (390×844)



DESKTOP (1440×900)

12 · Settings, Cloud Routing Selector & Scrollable In-App Provider Run Log

1. **On-Screen Functionality:** Host Voice selector (`Off`, `Chill`, `Full`), collapsible `⚙️ ADVANCED · Cloud routing` selector (`Baseten Primary → Fireworks Failover`, `Baseten Only`, `Fireworks AI Only`, `Dual Parallel`), scrollable in-app **Provider Run Log** (`30dvh` viewport showing verified run counts, cumulative tokens, model ID, score, and timestamp per call), `⚡ TEST & LOG TELEMETRY`, and `EXPORT LOG`.
2. **Value Prop:** Lets judges and operators trigger a live dual-provider `Muse Spark Open` verification in one tap and scroll the token audit trail directly inside the app.
3. **Tech Under the Hood:** `verifyLiveTelemetry()` invokes `callPartnerBackendProxy({ provider: "dual" })`, validates the returned JSON scorecards, and appends each provider's live token usage to `cleanquest.provider-log`.

LIVE · IN-APP TELEMETRY AUDIT

4 · Post-Hackathon MVP Roadmap — Features to Build & Unlock Next

Why These Were Scoped to Post-Hackathon: To keep the hackathon experience razor-focused for non-technical cleaners (Check-In → 1:30 Run → Scorecard → Daily Deck → Leaderboard & Export), multi-worker crew management, supervisor admin panes, physical QR sticker printing, chemical dwell timer credits, and Victory Pose emotes are held for the post-hackathon rollout. Several of these modules already have working data models and helper functions scaffolded in `app.js` (currently hidden via CSS scope trim) and are ready to unlock and expand next:

Post-Hackathon Feature	User & Business Value	Current Foundation in Code (`app.js`)	Next Engine Steps to Shi
1. Cleaner Benchmark Passport & 44px Glove PIN Identity *(MVP Screens 2 & 4)*	Gives cleaners a portable, shareable career credential with a **13-week consistency meter** (`Week N of 13 · On track` — never called a certification) and opt-in employer sharing, protected by a 4-digit glove-	`saveCleanerProfile`, `verifyCleanerPin`, `getConsistencyProgress`, `setEmployerShareOptIn`, and `shareCleanerPassport` are implemented and tested in `runPersonaSuite` (hidden via `#openPassportHubButton, #homePassportPanel`).	Unhide the Pa Hub card on Welcome/Rec signed sharea verification lir the serverless backend), an multi-device l

	friendly PIN pad with no email required.		
2. Physical Version-2 QR Station Stickers & In-App QR Scanner *(MVP Screen 5 & Admin Pane 12)*	Replaces dropdown picking with a 1-second wall-sticker scan at each sink, restroom, or classroom — proving physical presence at the exact station and unlocking prize eligibility automatically.	Standards-compliant Version-2 QR encoder (`encodeQrVersion2L`), payload parser (`parseAreaQrPayload` for `CQ-QR-<LEAGUE>-<AREA>-<SPOT>`), sticker printer (`printAreaQrSticker`), and `BarcodeDetector` modal (`openQrScannerModal`) are built in code.	Add cross-br `jsQR` canva for iOS WebK browsers lack `BarcodeDet` unhide `#scanAreaQ` and ship lami PDF sticker sl registered sit
3. Active +15s Chemical Dwell-Time Credit on Run Clock *(MVP Screen 6)*	Rewards cleaners for letting disinfectant sit for its required wet-contact time instead of rushing to wipe it dry — adding `+15s` dwell credit that never penalizes	`addDwellCredit()` and `state.checkin.dwellCreditSeconds` are wired into `timeBarExplanation()` and `computeTimeBenchmarkScore()` (button `#dwellCreditButton` hidden for hackathon simplicity).	Unhide `#dwellCueB` on stations th disinfectant c time, add visu spray confirm MediaPipe, ar dwell complia run.

	their speed score.		
4. Crews (3–8 Cleaners), Solo Virtual Squads & Weekly Matchups *(MVP Screen 4 & Admin Pane 13)*	Turns solitary night shifts into a team sport: 3–8 cleaners form a site crew (or solo cleaners join an automatic Virtual Squad), competing in weekly matchups with an 80% participation threshold and system-chosen Crew MVP.	`createCrew`, `joinCrew`, `createVirtualSquad`, `recordCrewRun`, `getCrewParticipation`, `getCrewMVP`, and `getWeeklyMatchup` are implemented and verified in the hermetic persona suite.	Move crew st browser `localStora` shared multi- storage on th backend prox teammates ac different phor live squad tot
5. Supervisor Non-Discipline Pledge, Team Training Gaps & AI Shift Briefings *(MVP Screens 11–15)*	Gives facility managers aggregate team training insights (top missed deck topics, shift time-bucket patterns) and 1-tap 5-day training pack assignments — gated behind a mandatory pledge that	`acceptAdminPledge`, `getTeamTrainingGaps` (strictly aggregate: `canEditScores: false`, `exposesIndividualMistakes: false`), `assignTrainingPack`, and `generateAdminAiBriefing` (`Muse Spark Open` with fingerprint cache) are coded in `#adminDashboardPanel`.	Build role-sep supervisor authenticatio cleaner log aggregation a site, and push assigned 5-d training packs devices.

	data is used for training and recognition only, never discipline.		
6. Victory Pose Silhouette Emotes (+5% Bonus) *(MVP Screen 8)*	Adds a playful post-run game-show finish: cleaners strike one of 4 silhouette emotes (`CHAMP V`, `FLEX`, `BOSS`, `DISCO`) on camera for up to `+5%` bonus points.	`loadPoseLandmarker` (MediaPipe 33-joint `Pose Landmarker Lite` + `GestureRecognizer`) and `computeStyleScore` are implemented in `app.js` (gate disabled and panel hidden during hackathon polish).	Refine front-camera/rear-c handoff after photo so clea strike a pose free with a 3- countdown ar cue.
7. Native Spotify & Apple Music OAuth + Automatic Voice Ducking *(Settings & Run Audio)*	Lets cleaners listen to their own personal playlists during a shift while CleanQuest automatically ducks music volume whenever the game-show host announces time callouts or verdicts.	Settings already includes zero-key Spotify/Apple Music embed drawer helpers (`renderMusicStreamEmbed`) and WebAudio gain ducking for the built-in tracks.	Integrate Spo Playback SDK 2.0 PKCE) an MusicKit JS v external strea playback duc smoothly und voice clips.

CleanQuest — Value Proposition, Feature Tour, Technical Architecture & Post-Hackathon MVP Roadmap · Launch Live App · Submission Playbook